# PG Lab-III

# Aim : Microwave Bridge Experiment

# for

# Dielectric Constant Measurement in Materials

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*Experiment performed on: Oct 6, 2025*

*Date of report submission: Oct 7, 2025*

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# 1 Abstract

Determining materials' dielectric constant (relative permittivity) is paramount in modern science and engineering applications. This report comprehensively investigates the Microwave Bridge experiment, a highly effective and widely employed technique for measuring the dielectric constant at microwave frequencies. The experiment uses a specialized mi- crowave bridge setup incorporating a microwave source, circulator (or isolator), directional coupler (or attenuator), and detector. The report discusses the calibration process, methodology, data analysis, and interpretation of results, highlighting the potential sources of error and explicitly estimating the errors.

# 2 Apparatus used

- Microwave source with Isolator, Attenuator, Frequency-counter, and Wave-guide
- Detector, Probe, and Multimeter
- Solid dielectric samples

# 3 Experimental Setup

## 3.1 Overview of the Microwave Bridge

The dielectric constant ($\varepsilon_r$) is an essential electrical property characterizing a material's response to an electric field. When subjected to electromagnetic fields, it determines the material's ability to store and transmit electrical energy. Accurate measurements of the dielectric constant are crucial for understanding material behavior and optimizing electronic and communication systems. Among various experimental techniques, the Microwave Bridge experiment stands out for its efficiency and precision in dielectric constant measurement at microwave frequencies. The Microwave Bridge experiment setup comprises a microwave source e.g., Gunn diode or Klystron tube (used in this experiment), circulator, directional coupler, and detector. The microwave source generates an electromagnetic signal that is transmitted through the material sample under test. The circulator and directional coupler isolate the incoming and outgoing signals, while the detector measures their power levels. The setup is care- fully calibrated to establish reliable reference points, allowing accurate dielectric constant measurements.

## 3.2 Sample Preparation

Before starting the experiment, the material sample needs to be prepared appropriately. For solid materials, the sample is usually cut into a specific shape and size to fit into the sample holder of the microwave bridge. For liquids, the sample is poured into a suitable container that can be placed in the microwave bridge. The sample dimensions and position within the bridge are crucial for accurate measurements. In this experiment, solid samples were used.

### 3.3 Calibration

Accurate calibration is the cornerstone of reliable dielectric constant measurements. The calibration involves two essential steps: open-circuit and short-circuit calibrations. In open- circuit calibration, the microwave bridge measures the reflection coefficient when the electro- magnetic signal encounters an open circuit. In short-circuit calibration, the bridge measures the reflection coefficient when the signal encounters a short circuit. These calibrations nullify cable and connector effects and ensure accurate dielectric constant determination.

## 4 Working Formulae

Assuming mono-chromatic plane wave solutions to Maxwell's equations in linear media, we can write

$$\nabla \times \vec{E} = -\partial_t \vec{B} = -j\omega\mu^* \vec{H}$$

and

$$\nabla \times \vec{H} = \partial_t \vec{D} + \sigma\vec{E} = j\omega\varepsilon^* \vec{E} + \sigma\vec{E}$$

where

$$\mu^* = \mu_0(\mu_r' - j\mu_r''), \quad \varepsilon^* = \varepsilon_0(\varepsilon_r' - j\varepsilon_r'')$$

The imaginary part of the complex dielectric permittivity ($\varepsilon^*$) is called the dielectric loss

$$\varepsilon_r'' \equiv \frac{\sigma}{\omega\varepsilon_0}$$

because it is responsible for the absorption of electromagnetic radiation whereas the real part i.e., the dielectric constant ($\varepsilon_r'$) is associated with the ability of a material to store electric energy. However, the dielectric constant is not independent of frequency but stays constant only over a small portion of the frequency spectrum. The slotted line section method uses the waveguide as the transmission line. The input impedance ($Z_i$) of the waveguide transmission line of length $l$, terminated by a load impedance, $Z_L$ is given by,

$$Z_i = Z_0 \frac{Z_L + jZ_0 \tan(\beta_{g0} l)}{jZ_L \tan(\beta_{g0} l) + Z_0}$$

where $Z_0$ is the characteristic impedance of the waveguide and $\beta_{g0} = \frac{2\pi}{\lambda_{g0}}$ is the propagation constant. If the waveguide is terminated with a short, then $Z_L = 0$ and therefore

$$Z_{i-\text{short}} = jZ_0 \tan(\beta_{g0} l)$$

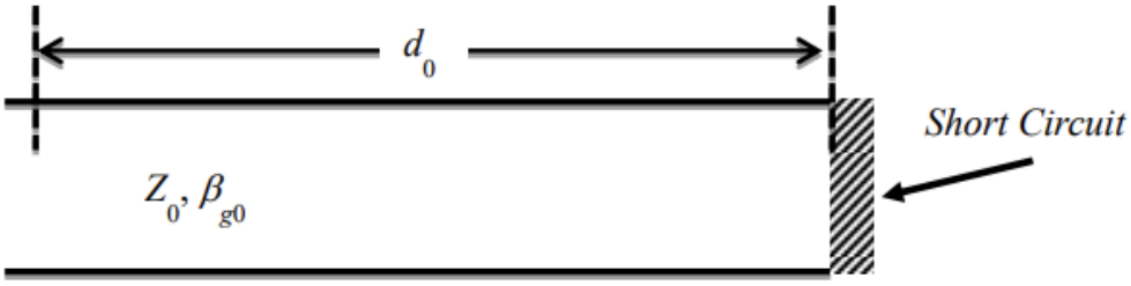



Figure 1: Showing the location of minimum at a distance of $d_0$ from the short-circuited end waveguide without sample

The voltage minimum observed by the slotted line detector indicates that the input impedance at that location, $Z_i = 0$. Therefore,

$$Z_i = 0 = \tan(\beta_{g0} d_0) \Rightarrow d_0 = \frac{n\lambda_{g0}}{2}$$

If a 'sample-short' combination terminates the waveguide, (say $Z_L = Z'_S$), then

$$Z_{i-\text{sample}} = Z_0 \frac{Z'_S + jZ_0 \tan \beta_{g0}(l-t)}{jZ'_S \tan \beta_{g0}(l-t) + Z_0}$$

where $t$ is the thickness of the sample backed by the short.

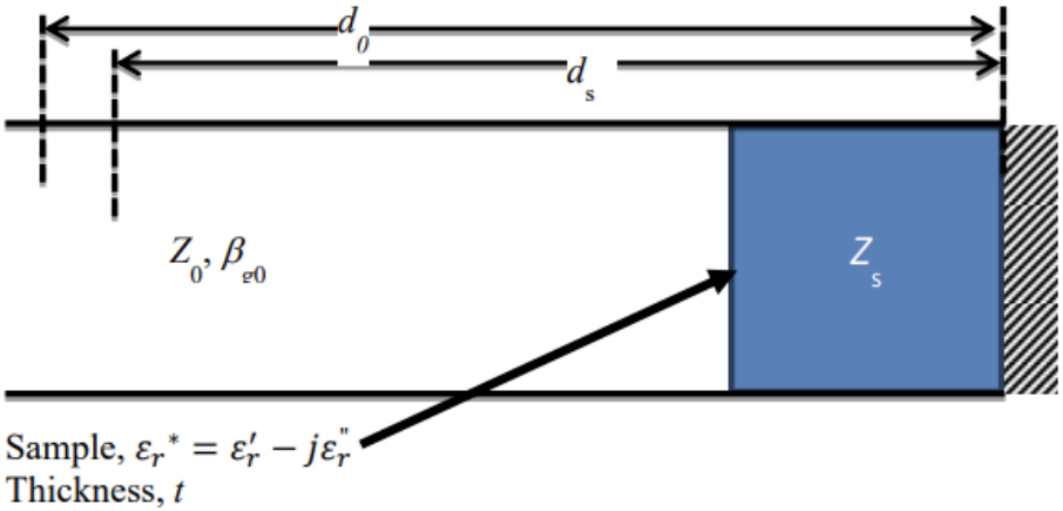



Figure 2: Showing the location of minimum at a distance of $d_s$ from the short-circuited end waveguide with sample

Again the voltage minimum observed by the slotted line detector indicates that the input impedance at that location, $Z_i = 0$. Therefore,

$$\begin{aligned} & Z'_s + jZ_0 \tan \beta_{g0}(d_s - t) = 0 \\ \Rightarrow\ & Z_s \tan(\beta_{gs} d_s) = -Z_0 \tan \beta_{g0}(d_s - t) \\ \Rightarrow\ & \frac{\tan \beta_{g0}(d_s - t)}{\tan(\beta_{gs} t)} = -\frac{Z_s}{Z_0} = \frac{\beta_{g0}}{\beta_{gs}} \end{aligned}$$

$$\Rightarrow \frac{\tan \beta_{g0}(t+x)}{\beta_{g0} t} = \frac{\tan(\beta_{gs} t)}{\beta_{gs} t}$$

where $d_s = d_0 + x$ and $\beta_{gs} = \frac{2\pi}{\lambda_{gs}}$.

So, the dielectric constant, $\varepsilon_r'$ can be evaluated using:

$$\varepsilon_r' = 1 + \frac{\beta_{gs}^2 - \beta_{g0}^2}{\beta_0^2}$$

where $\beta_0^2 = \beta_{g0}^2 + \frac{4\pi^2}{\lambda_c^2}$ and $\lambda_c = 2 \times$ (broad side of the waveguide)

# 5 Methodology

Once the calibration is complete, the sample is placed inside the microwave bridge, and the measurements can be taken. The bridge generates a microwave signal, which is transmitted through the sample. The material's dielectric properties influence the transmission and reflection of the microwave signal. By comparing the measured reflection coefficient or trans- mission coefficient with the calibrated values, the dielectric constant can be determined.

The procedure of the experiment is as follows:

1. The voltage is plotted as a function of the position of the probe using the slotted line section, with no sample in the short-circuited line.
2. Half of the wavelength of the microwave in the waveguide ($\lambda_{g0}/2$) is evaluated from the distance between the two adjacent minima positions.
3. The values of $d_0$ are noted down from the plot of voltage versus the position of the probe.
4. The end plate (acting as a short) is opened, and the sample is inserted. Then, the end plate is replaced in such a manner that it firmly attaches to the end of the sample.
5. With the sample, the position of the minima in the slotted line section is changed to $d_s$. If two or three positions for $d_0$ were taken, corresponding positions for $d_s$ may be taken and tabulated for a given sample.
6. Previous steps are repeated with a sample having a different thickness.
7. The shift in the minima with and without the sample is calculated by $|d_0 - d_s|$.
8. The dielectric constant is evaluated using the working formulae.

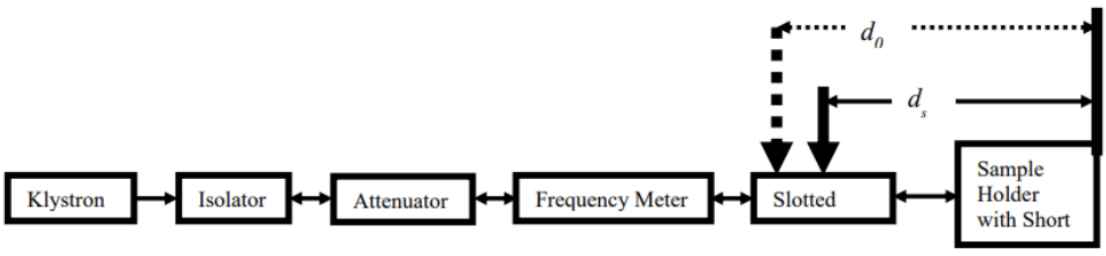



Figure 3: The whole setup

# 6 Results

## 6.1 For Air

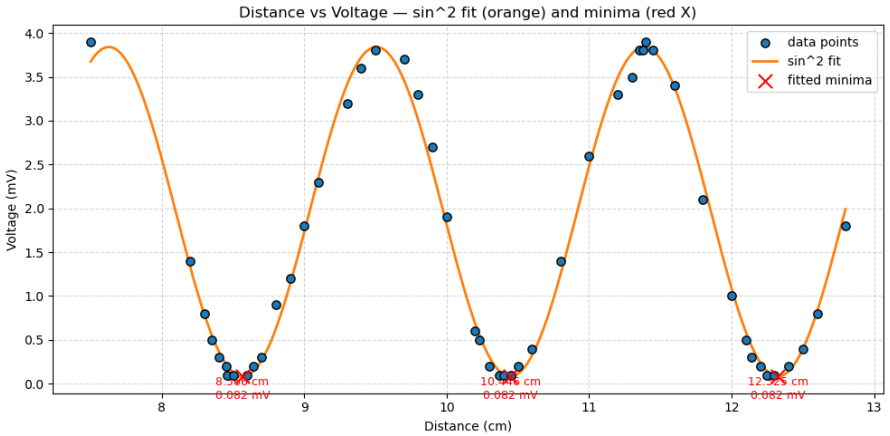



Figure 4: Plot for the Air

Table 1: Multimeter reading vs Position for the waveguide without sample

| Distance (cm) | Multimeter Reading (mV) | spacing (cm) |
|---|---|---|
| 8.56 | 0.1 | - |
| 10.45 | 0.1 | 1.89 |
| 12.32 | 0.1 | 1.87 |

## 6.2 For Glass sample

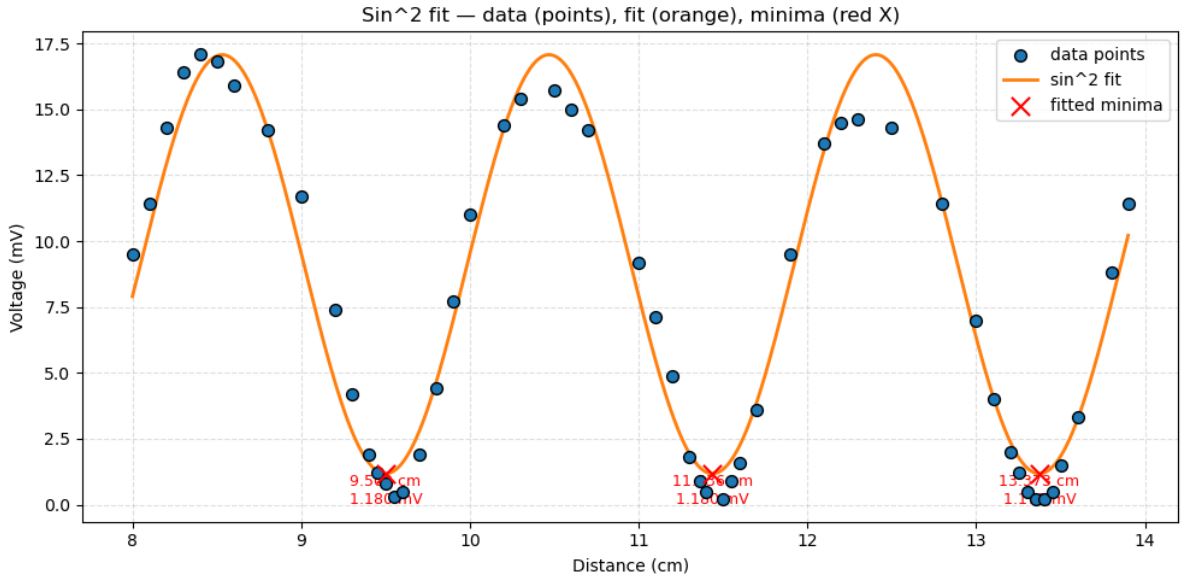



Figure 5: plot for glass sample

Table 2: To determine $\beta_{g0}$ for the waveguide with the glass sample

| Locations of minima (cm) | Differences between minima (cm) | Avg. of the differences (cm) | $\lambda_{g0} = 2\bar{d}$ (cm) | $\beta_{g0} = \frac{2\pi}{\lambda_{g0}}$ (cm$^{-1}$) |
|---|---|---|---|---|
| 9.5 | | | | |
| 11.44 | 1.94 | | | |
| 13.37 | 1.93 | 1.935 | 3.87 | $\approx 1.624$ |

Given that, the broad side of the waveguide = 2.23 cm. So, $\lambda_c = 2 \times 2.23$ cm $= 4.46$ cm. Thus

$$\beta_0^2 = (1.624)^2 + \frac{4\pi^2}{(4.46)^2}\,\text{cm}^{-2} \approx 4.62\,\text{cm}^{-2}$$

Next, we find the $x = |d_s - d_0|$ values corresponding to each minima which turn out to be [0.95, 0.88] respectively and so their avg. value, $\bar{x} \approx 0.915$ cm. The thickness of the sample was found to be 1 cm. Substituting the values into eqn.(6), we get:

$$\frac{\tan(\beta_{gs}t)}{\beta_{gs}t} = \frac{\tan[1.624 \times (1 + 0.915)]}{1.624 \times 1}$$

and solving this transcendental equation in Python, $\beta_{gs}t \approx 3.08$.
Putting all the values in eqn. (7), we have $\varepsilon_r' \approx 2.48$.

## 6.3 For teflon sample

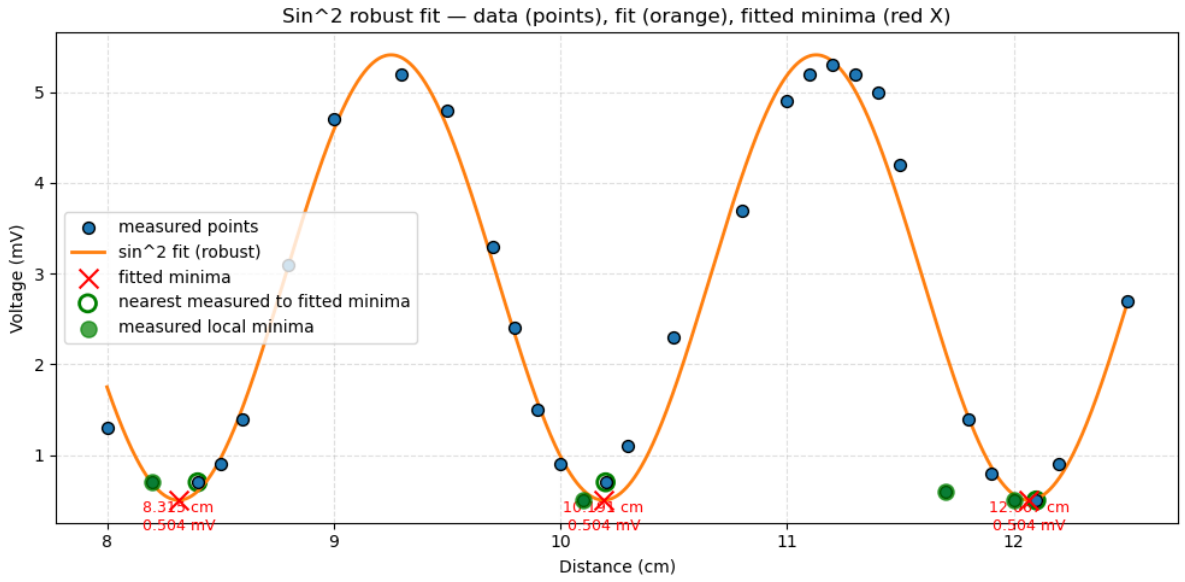



Figure 6: plot for teflon

Table 3: To determine $\beta_{g0}$ for the waveguide with the teflon sample

| Locations of minima (cm) | Differences between minima (cm) | Avg. of the differences (cm) | $\lambda_{g0} = 2\bar{d}$ (cm) | $\beta_{g0} = \frac{2\pi}{\lambda_{g0}}$ (cm$^{-1}$) |
|---|---|---|---|---|
| 8.32 | | | | |
| 10.19 | 1.87 | | | |
| 12.06 | 1.87 | $\approx 1.87$ | $\approx 3.74$ | $\approx 1.68$ |

Now, we find the $x = |d_s - d_0|$ values corresponding to each minimum, which are [0.24, 0.26, 0.26] respectively and so their avg. value, $\bar{x} \approx 0.253$ cm. The thickness of the sample was found to be 0.93 cm. Substituting the values into eqn.(6), we get:

$$\frac{\tan(\beta_{gs}t)}{\beta_{gs}t} = \frac{\tan[1.68 \times (0.93 + 0.26)]}{1.68 \times 0.93}$$

$$\beta_0^2 = (1.68)^2 + \frac{4\pi^2}{(4.46)^2}\,\mathrm{cm}^{-2} \approx 4.807\,\mathrm{cm}^{-2}$$

and solving this transcendental equation in Python, $\beta_{gs}t \approx 1.917$.

Putting all the values in eqn. (7), we have $\varepsilon_r' \approx 1.29$.

# 7 Conclusion

The Microwave Bridge experiment is a powerful and widely adopted technique for measuring the dielectric constant of materials at microwave frequencies. Its comprehensive calibration process, accurate methodology, and profound data analysis make it an indispensable tool in modern materials science research. In this experiment, we have obtained the dielectric constants of the given samples to be slightly different but we expect them to be the same because both the samples are made of the same material, and the microwave generator was kept at the same frequency throughout the experiment.